Responses to Reviewers' Comments for Manuscript JSAC-01286-2025

Decentralized Parameter-Free Online Learning

Addressed Comments for Publication to

IEEE Journal on Selected Areas in Communications

by

Tomas Ortega and Hamid Jafarkhani

Dear Dr. Zhiguo Ding,

Please find enclosed the revised version of our previous submission entitled "Decentralized Parameter-Free Online Learning" with manuscript number JSAC-01286-2025. We would like to thank you and the reviewers for the valuable comments which helped improving the quality of our manuscript. In this revision, we have carefully addressed the reviewers' comments. A summary of main modifications and a detailed point-by-point response to the comments from Reviewers 1 to 3 (following the reviewers' order in the decision letter) are given below.

Sincerely,

Tomas Ortega and Hamid Jafarkhani

Note: To enhance the legibility of this response letter, all the editor's and reviewers' comments are typeset in boxes. Rephrased or added sentences are typeset in red. The respective parts in the manuscript are blue to indicate changes.

Contents

Authors' Response to the Editor

General Comments. The manuscript requires a major revision. The main concerns are communication cost in constrained decentralized networks, the gap between the theoretical increasing-gossip schedule and practical constant communication, the role of DECO-ii relative to DECO-i, and the need for stronger empirical comparisons including adaptive baselines and real-data experiments.

Response: We thank the editor and reviewers for the detailed and constructive comments.

We revised the manuscript and repository to address these concerns directly. The main changes are:

1. We made the communication model explicit: one gossip round is one local exchange over each directed neighbor relation. We now count communicated real scalars and distinguish DECO-i, which sends (Wealth, G), from DECO-ii, DOGD, and the three adaptive baselines, which send one d-dimensional vector.
2. We clarified that $q(t) = \mathcal{O}(t)$ is a sufficient schedule for the current proof, not a claim that quadratic communication is necessary or desirable in constrained networks. The practical experiments emphasize the constrained $q(t) = 1$ regime.
3. We reframed DECO-ii as a one-state betting-function formulation rather than an unqualified approximation to DECO-i. We explicitly state the communication/performance trade-off and do not claim that DECO-ii uniformly improves over DECO-i.
4. We added three representative online decentralized adaptive baselines—D-AdaGrad, D-RMSProp, and D-Adam—with the same gossip interface and communication accounting as DOGD.
5. We swept DOGD and all three adaptive baselines over the same 25 learning-rate scales on identical synthetic and LIBSVM online streams, and included them in every experimental figure and the real-data table.

6. We revised the experimental discussion to avoid overclaiming empirical dominance: the best-in-grid adaptive methods can be competitive or better on some streams, while the central contribution of DECO is avoiding learning-rate selection while retaining network-regret guarantees and explicit communication accounting.

For transparency, the exact added or modified manuscript text supporting every item is reproduced under the corresponding point-by-point response below; no separate manuscript change was made solely for this summary.

Authors' Response to Reviewer 1

General Comments. The paper proposes a family of parameter-free decentralized online learning algorithms based on coin-betting and a proof of sublinear regret. The reviewer asks for a clearer communication setup, the gossip overhead, a more careful discussion of the centralized gap, and intuition for a real application.

Response: We thank the reviewer for identifying the communication-efficiency issues that needed to be stated more concretely.

Comment 1

Although the paper proposes an interesting solution backed by regret guarantees, it is not clear what communication setup is considered or how the approach fits constrained networks.

Response: Thank you for the constructive comment.

We revised the manuscript to explicitly describe the communication setup. The network is an undirected connected graph, implemented through a Metropolis-Hastings gossip matrix, and one gossip round means that each agent exchanges its current communicated state with its immediate neighbors only. The manuscript now states:

> We assume synchronous exchanges over a fixed, connected, undirected graph and construct W with Metropolis-Hastings weights in the experiments. One gossip round consists of one local exchange over every directed neighbor relation induced by the graph; no all-to-all exchange or central coordinator is used. If d is the model dimension and $|E_{\mathrm{dir}}|$ is the number of directed non-self neighbor transmissions, one gossip round communicates $|E_{\mathrm{dir}}|d$ real scalars for DECO-**ii**, DOGD, and the adaptive decentralized baselines, and $|E_{\mathrm{dir}}|(d+1)$ real scalars for DECO-**i** because it additionally communicates the scalar wealth. Over a horizon T, a schedule

$q(t)$ therefore communicates $|E_{\mathrm{dir}}|d \sum_{t=1}^{T} q(t)$ scalars for one-vector methods and $|E_{\mathrm{dir}}|(d+1) \sum_{t=1}^{T} q(t)$ scalars for DECO-**i**. Consequently, $q(t) = 1$ has linear total communication in T, whereas $q(t) = \lceil \alpha t \rceil$ has quadratic total communication.

This makes clear that the proposed method is local-neighbor decentralized and that the communication cost is now measurable per algorithm and per graph.

Comment 2

How is gossiping happening? What is the communication overhead? The paper states that $O(t)$ gossiping steps are needed in each round, implying $O(T^2)$ gossiping over the horizon.

Response: Thank you for the constructive comment.

We agree that the previous text blurred a sufficient condition for the proof with a practical communication recommendation. We now state that $q(t) = \mathcal{O}(t)$ is sufficient for the current disagreement proof, but the practical constrained-network regime is constant communication, $q(t) = 1$. For a horizon T, constant gossip has cost $\mathcal{O}(T|E_{\mathrm{dir}}|d)$ for one-vector methods and $\mathcal{O}(T|E_{\mathrm{dir}}|(d+1))$ for DECO-i; a linear schedule has $\mathcal{O}(T^2|E_{\mathrm{dir}}|d)$ cost. We added code-level accounting of total communicated scalars and report this in the new review experiments. The exact revised manuscript text is:

Over a horizon T, a schedule $q(t)$ therefore communicates $|E_{\mathrm{dir}}|d \sum_{t=1}^{T} q(t)$ scalars for one-vector methods and $|E_{\mathrm{dir}}|(d+1) \sum_{t=1}^{T} q(t)$ scalars for DECO-**i**. Consequently, $q(t) = 1$ has linear total communication in T, whereas $q(t) = \lceil \alpha t \rceil$ has quadratic total communication.

They establish that by performing $\mathcal{O}(t)$ gossip steps at each round, the per-round communication grows linearly and the total communication through horizon T

grows as $\mathcal{O}(T^2)$; this sufficient schedule controls the disagreement term to the order of the local regret term and ensures sublinear network regret.

The schedule $q(t) = \mathcal{O}(t)$ is sufficient under the present disagreement analysis and is not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment illustrates the mechanism covered by the current proof.

Comment 3

The claim that a linear schedule closely tracks the centralized oracle is too strong because the figure shows a substantial gap.

Response: Thank you for the constructive comment.

We agree. We revised the discussion to treat the centralized gap as a real finite-horizon decentralization cost rather than as a negligible discrepancy. The exact revised discussion is:

For $q(t) = \lceil 0.1t \rceil$, the cumulative number of gossip rounds is quadratic in T, whereas the constant schedule uses only T rounds. The linear schedule reduces loss relative to the constant and logarithmic schedules, but a visible finite-horizon gap to the centralized oracle remains. This gap is a real decentralization cost rather than a claim of equality with the oracle. The linear schedule should be interpreted as a theory-motivated sufficient schedule for controlling the current disagreement bound, not as the recommended operating point for every constrained network.

The figure also includes D-AdaGrad, D-RMSProp, and D-Adam with $q(t) = 1$ and $\eta_0 = 0.1$ as online adaptive reference curves; only DECO is varied across gossip schedules.

Revised figure caption: Cumulative network loss on the synthetic online stream: DECO-**ii** (KT) is shown with constant, logarithmic, and linear gossip schedules; D-AdaGrad, D-RMSProp, and D-Adam use $q(t) = 1$ and $\eta_0 = 0.1$; increasing DECO's communication budget reduces loss, while the adaptive curves provide same-communication references.

Thus, the revised discussion no longer relies on an unqualified "closely tracks" interpretation.

Comment 4

Intuition into how everything is tied to a real application is missing.

Response: Thank you for the constructive comment.

We expanded the operational description in the manuscript to connect the algorithm to distributed sensing and collaborative prediction. Each node makes a local prediction, observes local feedback, updates its coin-betting state, and then gossips only with physical or logical neighbors. This matches applications such as sensor networks, device-to-device learning, or collaborative streaming regression, where no central coordinator is available and the communication budget is dominated by local neighbor exchanges. The added manuscript text is:

An operational example is collaborative streaming regression over a sensor or device-to-device network. At round t, each node predicts from its local stream, observes its local feedback, updates its coin-betting state, and gossips the communicated state only with physical or logical neighbors. No central coordinator is required, and the communication budget is determined by these local neighbor exchanges.

Authors' Response to Reviewer 2

General Comments. The reviewer raises concerns about novelty, theory depth, the theory/practice gap caused by increasing gossip rounds, and limited experiments and baselines.

Response: We thank the reviewer for pushing us to sharpen the contribution and empirical claims.

Comment 1

The claimed novelty is not sufficiently justified; the method combines coin-betting and gossip and may appear to be a straightforward extension.

Response: Thank you for the constructive comment.

We revised the positioning to be more precise. We no longer frame the contribution as novelty from gossip alone. Instead, we emphasize three specific contributions: (i) a decentralized online-learning formulation with network regret guarantees for parameter-free coin-betting agents; (ii) an explicit decomposition of regret into average local coin-betting regret and a communication-controlled disagreement term; and (iii) the betting-function formulation underlying DECO-ii, which permits a one-state gossiped update and a direct network-regret analysis. We also expanded the related-work discussion to distinguish our convex online-learning setting from decentralized stochastic/nonconvex parameter-free optimization and from methods that require stepsizes or different problem models. The manuscript now states:

> Our contribution is not a new gossip operator, reward-regret duality, or coin betting in isolation. It is the decentralized online-learning formulation and analysis that (i) equips parameter-free coin-betting agents with network-regret guarantees, (ii) decomposes network regret into average local coin-betting regret and a

communication-controlled disagreement term, and (iii) uses the one-state betting-function formulation of DECO-**ii** to obtain a direct network-regret analysis.

Recent parameter-free decentralized optimization methods address nonconvex stochastic objectives [17], strongly convex smooth objectives [18], and composite convex objectives [19]. These works study convergence to a minimizer under their respective optimization models, whereas our setting considers sequential convex losses and network regret against a comparator. The problem models and performance criteria are therefore complementary rather than directly interchangeable.

Comment 2

The theoretical contribution appears limited and the proof techniques resemble existing reward-regret duality and gossip averaging arguments.

Response: Thank you for the constructive comment.

We agree that the building blocks are classical, and the revised text is more explicit about this. The contribution is in how these tools are combined for network regret in a decentralized online setting, and in identifying the disagreement term that must be controlled when coin-betting decisions are driven by gossiped cumulative gradients. We clarified the proof structure so that readers can see which parts follow standard coin-betting duality and which parts are specific to the decentralized state and betting-function formulation. The added manuscript text is:

The analysis combines two classical ingredients: reward-regret duality for the average local coin-betting term and gossip-mixing bounds for consensus. The decentralized component is their coupling when decisions are generated from gossiped cumulative gradients. In particular, the disagreement term exposes how the betting-function Lipschitz constants, network spectral radius, and gossip schedule jointly affect network regret.

Comment 3

The theory relies on increasing communication, while the experiments primarily use constant communication, creating a theory/practice gap.

Response: Thank you for the constructive comment.

We agree and now state this explicitly. The sufficient $q(t) = \mathcal{O}(t)$ schedule is used to prove sublinear disagreement under the current analysis. However, it is not presented as a practical requirement. The constant-communication experiments address the constrained-network regime, while the increasing-communication experiment illustrates the theoretical mechanism. We added text explaining that closing the theory/practice gap for $q(t) = \mathcal{O}(1)$ remains an important theoretical question. The manuscript now states:

They establish that by performing $\mathcal{O}(t)$ gossip steps at each round, the per-round communication grows linearly and the total communication through horizon T grows as $\mathcal{O}(T^2)$; this sufficient schedule controls the disagreement term to the order of the local regret term and ensures sublinear network regret.
The schedule $q(t) = \mathcal{O}(t)$ is sufficient under the present disagreement analysis and is not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment illustrates the mechanism covered by the current proof.

Comment 4

The experimental evaluation is not sufficiently convincing; baselines are limited and the experiments should include adaptive/parameter-free methods, heterogeneous data, larger networks, or adversarial settings.

Response: Thank you for the constructive comment.

We added three representative online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam. They cover cumulative second-moment adaptation, exponentially weighted second-moment adaptation, and combined first/second-moment adaptation, respectively. We did not add AdamW as a near-duplicate of Adam with weight decay, or momentum/Nesterov as additional non-adaptive momentum controls; the selected three directly address the reviewer's request for adaptive-gradient comparisons without making the figures unreadable. Every baseline follows the same causal predict-observe-update-gossip protocol and is evaluated on an identical stream. We ran DOGD and all three adaptive methods at the same 25 initial scales from 10^{-3} to 10^{3}. The complete sweeps appear in the synthetic and four-dataset figures, while the connectivity and gossip figures include all three adaptive methods at one clearly labeled reference setting. The real-data table reports the measured best-in-grid result and communication count for every method. The exact added or modified manuscript text, including all affected figure captions and the complete table, is reproduced below in manuscript order:

We compare them against the following baselines:

Online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam. These three representatives respectively use cumulative second moments, exponentially weighted second moments, and combined first/second moments.

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^{3}]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses

$\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. The figures report cumulative network loss over the complete sweep; the table reports the best value in hindsight over that displayed grid and therefore does not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the common base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. DOGD, D-AdaGrad, D-RMSProp, and D-Adam are swept over all 25 values; the horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^3 on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the sensitivity figure and the remaining comparisons show the KT potential for DECO and the centralized reference to keep the seven-curve legends readable.

For context, the same figure also reports D-AdaGrad, D-RMSProp, and D-Adam at the representative setting $p = 0.3$, $q(t) = 1$, and $\eta_0 = 0.1$; these reference curves are not additional connectivity sweeps.

Synthetic online performance under varying connectivity. The ER curves sweep DECO-**ii** (KT) over $p \in \{0.1, 0.3, 1.0\}$; D-AdaGrad, D-RMSProp, and D-Adam are $p = 0.3$, $q(t) = 1$, $\eta_0 = 0.1$ references. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

The figure also includes D-AdaGrad, D-RMSProp, and D-Adam with $q(t) = 1$ and $\eta_0 = 0.1$ as online adaptive reference curves; only DECO is varied across gossip schedules.

Cumulative network loss on the synthetic online stream: DECO-**ii** (KT) is shown with constant, logarithmic, and linear gossip schedules; D-AdaGrad, D-RMSProp, and D-Adam use $q(t) = 1$ and $\eta_0 = 0.1$; increasing DECO's communication budget reduces loss, while the adaptive curves provide same-communication references.

Final cumulative network loss on four online LIBSVM streams. Each panel shows the complete 25-value η_0 sweep for DOGD, D-AdaGrad, D-RMSProp, and D-Adam on a log-log scale. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^3]$. Table II reports the best cumulative network loss in hindsight over the displayed grid, together with the selected η_0 and total communicated scalars.

Online LIBSVM experiments with $N = 20$ and $q(t) = 1$. For DOGD and the three adaptive baselines, "Selected η_0" is the best value in hindsight among the 25 values plotted in Fig. 6; it is not additional tuning hidden from the figure. Lower cumulative network loss is better.

Dataset	T	Method	Selected η_0	Cumulative network loss	Scalars
space_ga	155	DECO-i (KT)	—	92.51	43,400
space_ga	155	DECO-ii (KT)	—	101.76	37,200
space_ga	155	DOGD	1	88.81	37,200
space_ga	155	D-AdaGrad	0.316	89.21	37,200
space_ga	155	D-RMSProp	0.0316	89.28	37,200
space_ga	155	D-Adam	0.178	89.53	37,200
cpusmall	409	DECO-i (KT)	—	136.59	212,680
cpusmall	409	DECO-ii (KT)	—	147.13	196,320
cpusmall	409	DOGD	1.78	127.63	196,320
cpusmall	409	D-AdaGrad	0.562	121.73	196,320
cpusmall	409	D-RMSProp	0.0316	122.99	196,320
cpusmall	409	D-Adam	0.1	122.09	196,320
cadata	1,032	DECO-i (KT)	—	517.87	371,520
cadata	1,032	DECO-ii (KT)	—	583.69	330,240
cadata	1,032	DOGD	1.78	495.55	330,240
cadata	1,032	D-AdaGrad	1	493.68	330,240
cadata	1,032	D-RMSProp	0.0562	496.91	330,240
cadata	1,032	D-Adam	0.1	499.85	330,240
abalone	208	DECO-i (KT)	—	130.13	74,880
abalone	208	DECO-ii (KT)	—	137.66	66,560
abalone	208	DOGD	1.78	121.45	66,560
abalone	208	D-AdaGrad	0.562	121.30	66,560
abalone	208	D-RMSProp	0.0562	121.50	66,560
abalone	208	D-Adam	0.1	121.07	66,560

The best-in-grid adaptive baselines are competitive with or better than DECO on several streams, but the selected scale varies by method and dataset. The empirical conclusion is therefore not that DECO uniformly dominates tuned adaptive methods; rather, DECO avoids this learning-rate sweep while providing explicit network-regret guarantees in the convex online setting.

Finally, we evaluate DECO, DOGD, and the three online adaptive baselines on four real-world LIBSVM regression streams. The cycle graph, sample order, horizon, and one-gossip-round budget are identical for every method and every value of η_0. Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, as the best-in-grid values in Table II make explicit. The supported conclusion is that DECO removes learning-rate selection, not that it uniformly improves upon every tuned adaptive optimizer.

The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.

Authors' Response to Reviewer 3

General Comments. The reviewer asks for notation cleanup, a quantitative explanation of DECO-ii relative to DECO-i, clarification of whether DECO-ii satisfies the needed coin-betting condition, discussion of potential-function choices, adaptive baselines, related parameter-free decentralized work, limitations to convex online learning, and fair communication-normalized DECO-i/DECO-ii comparisons.

Response: We thank the reviewer for the detailed technical comments.

Comment 1

The notation is inconsistent: both g and $c = -g$ are used, and c is overloaded later.

Response: Thank you for the constructive comment.

We revised the notation discussion to reduce overloading. The manuscript now emphasizes $g_{n,t}$ as the subgradient feedback and uses the identification $c_t = -g_t$ locally when recalling the classical single-agent coin-betting view. We also avoid reusing c for unrelated constants in the disagreement analysis and corrected the operational update to match the algorithm's $G - g$ convention. This makes the OCO and coin-betting interpretations easier to follow. The exact added notation text and corrected operational equation are:

To keep the two viewpoints distinct, $g_{n,t}$ denotes subgradient feedback throughout the OCO and network-regret development, whereas $c_t = -g_t$ is used only in the local recap of the classical coin-betting game. The coefficient of a linear gossip schedule is denoted by κ, so $q(t) = \lceil \kappa t \rceil$; it is not a coin outcome.

$$\hat{G}_{n,t} = G_{n,t-1} - g_{n,t}.$$

Every $c \to \kappa$ modification in the two schedule theorems and proofs is reproduced below in manuscript order; only unchanged intervening text is omitted:

and choosing the gossip step function to be

$$q(t) = \lceil \kappa \cdot t \rceil,$$

for any $\kappa \geq \frac{-3}{2 \ln \rho}$ has sublinear regret and disagreement.

Now, let $q(t) = \kappa \cdot t$. Notice that we can express

$$Q(s,t) = \kappa \cdot \frac{(t-s+1)(s+t)}{2},$$

Thus, we can upper bound

$$e^{t/2} \sum_{s=1}^{t-1} \rho^{Q(s,t)} \leq e^{t/2}(t-1)\rho^{\kappa(2t-1)}.$$

We want to ensure that

$$e^{t/2}(t-1)\rho^{\kappa(2t-1)} \leq 1$$

for all $t \geq 2$. Taking logarithms, we obtain

$$\frac{t}{2} + \ln(t-1) + \frac{\kappa \ln \rho}{2}(2t-1) \leq 0$$

The derivative of the LHS is

$$\frac{1}{2} + \frac{1}{t-1} + \kappa \ln(\rho),$$

and ensuring

$$\kappa \geq \frac{-3}{2\ln(\rho)},$$

it is non-positive for all $t \geq 2$, which ensures the non-increasing property.

and choosing the gossip step function to be

$$q(t) = \lceil \kappa \cdot t \rceil,$$

for any $\kappa \geq \frac{-2\ln(2)}{\ln(\rho)}$ has sublinear regret and disagreement.

We set $q(t) = \kappa \cdot t$ and check that

$$2^t(\ln t - \psi(1/2))(t-1)\rho^{\kappa(2t-1)} \leq 1.$$

For $t \geq 2$, we can take logarithms and obtain

$$t\ln(2) + \ln(\ln t - \psi(1/2)) + \ln(t-1) + \frac{\kappa \ln \rho}{2}(2t-1) \leq 0.$$

for all $t \geq 2$, and the condition simplifies to

$$2t\ln(2) + \frac{\kappa \ln \rho}{2}(2t-1) \leq 0,$$

which is satisfied when $\kappa \geq \frac{-2\ln(2)}{\ln(\rho)}$.

Comment 2

DECO-ii seems to replace Wealth_{t-1} by $F_{t-1}(G_{t-1})$. The paper should quantify the approximation error and its regret effect.

Response: Thank you for the constructive comment.

We clarified that DECO-ii should not be read as a heuristic approximation to DECO-i with an unquantified error term. It is a one-state betting-function formulation with update $x_{n,t} = h_t(G_{n,t-1})$. The analysis for DECO-ii is stated directly in terms of the betting function and the disagreement induced by gossiped cumulative gradients. Empirically, we also acknowledge that DECO-i can perform better, and that DECO-ii trades off some performance for a smaller communicated state. The manuscript now states:

DECO-**ii** is defined directly by the one-state update $x_{n,t} = h_t(G_{n,t-1})$; it is not obtained by substituting the approximation $\text{Wealth}_{n,t-1} \approx F_{t-1}(G_{n,t-1})$ into DECO-**i**. Its regret analysis is therefore stated in terms of h_t and the disagreement induced by gossiped cumulative gradients, rather than an approximation-error term relative to DECO-**i**. This smaller communicated state can trade performance for communication, and DECO-**ii** is not claimed to dominate DECO-**i**.

Comment 3

Lemma 3 verifies the condition for DECO-i, but the paper should verify whether DECO-ii satisfies the same condition.

Response: Thank you for the constructive comment.

We added a direct verification for the betting-function update, rather than relying on the omitted-proof statement in the previous manuscript. The new derivation applies the potential condition before gossip, uses convexity and double stochasticity after gossip, and telescopes over time. It shows explicitly that DECO-ii obtains the potential inequality needed by reward-regret duality without storing or gossiping wealth. We also revised the average-local-regret proof so that it invokes the appropriate argument for each variant. The complete added and modified manuscript text is:

Direct potential verification for DECO-ii. For DECO-**ii**, let $\beta_{n,t} = \beta_t(G_{n,t-1})$ and $x_{n,t} = h_t(G_{n,t-1}) = \beta_{n,t} F_{t-1}(G_{n,t-1})$. Substituting the coin outcome $-g_{n,t}$ in condition (c) of the coin-betting potential gives

$$\begin{aligned} F_{t-1}(G_{n,t-1}) - g_{n,t} x_{n,t} &= (1 - g_{n,t}\beta_{n,t})\, F_{t-1}(G_{n,t-1}) \\ &\geq F_t(G_{n,t-1} - g_{n,t}) = F_t(\widehat{G}_{n,t}). \end{aligned}$$

After gossip, $G_{n,t} = \sum_m W_{mn} \widehat{G}_{m,t}$, so convexity of F_t gives

$$\begin{aligned} F_t(G_{n,t}) &\leq \sum_m W_{mn} F_t(\widehat{G}_{m,t}) \\ &\leq \sum_m W_{mn} \left(F_{t-1}(G_{m,t-1}) - g_{m,t} x_{m,t} \right). \end{aligned}$$

Summing over agents and using double stochasticity yields

$$\sum_n F_t(G_{n,t}) \leq \sum_n F_{t-1}(G_{n,t-1}) - \sum_n g_{n,t} x_{n,t}.$$

Telescoping from $F_0(0) = \epsilon$ and applying Jensen's inequality therefore gives

$$\begin{aligned}
\frac{1}{N}\sum_{t=1}^{T}\sum_{n=1}^{N} g_{n,t} x_{n,t} &\leq \epsilon - \frac{1}{N}\sum_{n=1}^{N} F_T(G_{n,T}) \\
&\leq \epsilon - F_T\left(\frac{1}{N}\sum_{n=1}^{N} G_{n,T}\right) \\
&= \epsilon - F_T\left(-\sum_{t=1}^{T} \bar{g}_t\right).
\end{aligned}$$

Thus, DECO-**ii** satisfies the potential inequality needed by reward-regret duality without storing or gossiping a wealth variable.

Using Lemma 3 for DECO-**i** and the direct potential verification above for DECO-**ii**, we can derive regret bounds for both algorithms.

For DECO-**i**, the algorithmic state $G_{n,t}$ equals the accumulated coin outcome $C_{n,t}$ because both start at zero, subtract the same subgradient locally, and undergo the same gossip update. Hence, the wealth lower bound in Lemma 3 and preservation of network averages give the required reward inequality. For DECO-**ii**, the direct potential verification above gives the same inequality without introducing a wealth variable. Thus, for either variant,

$$\begin{aligned}
\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T}\langle g_{n,t}, x_{n,t}\rangle &\leq \epsilon - \frac{1}{N}\sum_{n=1}^{N} F_T(G_{n,T}) \\
&\leq -F_T\left(-\sum_{t=1}^{T} \bar{g}_t\right) + \epsilon.
\end{aligned}$$

Comment 4

The method removes learning-rate tuning but introduces the choice of potential function and storage of cumulative gradients.

Response: Thank you for the constructive comment.

We agree and revised the language to avoid implying that all design choices disappear. The method is learning-rate-free and horizon/comparator adaptive, but the potential function is still a modeling choice. We now state this explicitly and report both KT and exponential potentials. We also clarify that cumulative-gradient storage is one d-dimensional vector per agent, comparable to storing optimizer state in adaptive-gradient methods. The added manuscript text is:

> Here parameter-free means learning-rate-free and comparator/horizon adaptive; it does not mean free of every design choice. The potential family remains a modeling choice, and we evaluate both KT and exponential potentials. Each DECO-**ii** agent stores one d-dimensional accumulated-gradient vector; DECO-**i** additionally stores one scalar wealth. This memory is comparable to the vector state used by adaptive-gradient optimizers.

Comment 5

The evaluation should include adaptive methods such as AdaGrad, RMSProp, Adam, AdamW, momentum, and Nesterov-type methods.

Response: Thank you for the constructive comment.

We selected D-AdaGrad, D-RMSProp, and D-Adam as three representative adaptive-gradient baselines. Respectively, they cover cumulative second moments, exponentially weighted second moments, and combined first/second moments. AdamW mainly adds weight decay to Adam, while momentum and Nesterov are not adaptive-gradient methods;

using these three representatives keeps the seven-curve figures legible while directly testing the reviewer's adaptive-method concern. We implemented each method as a genuinely online decentralized algorithm: at every round it predicts, observes only the current local gradient, updates its local optimizer state, and gossips one decision vector. DOGD and all three adaptive methods use the same 25-value initial-rate grid and identical streams. We ran the complete synthetic and four-LIBSVM sweeps, added all three methods to every experimental figure, and report their measured results and communication counts in Table II. The exact added or modified manuscript text, including all affected figure captions and the complete table, is:

We compare them against the following baselines:

Online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam. These three representatives respectively use cumulative second moments, exponentially weighted second moments, and combined first/second moments.

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^{3}]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. The figures report cumulative network loss over the complete sweep; the table reports the best value in hindsight over that displayed grid and therefore does not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the common base-rate grid $\eta_0 \in [10^{-3}, 10^{3}]$. DOGD, D-AdaGrad, D-RMSProp, and D-Adam are swept over all 25 values; the horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^{3} on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the sensitivity figure and the remaining comparisons show the KT potential for DECO and the centralized reference to keep the seven-curve legends readable.

For context, the same figure also reports D-AdaGrad, D-RMSProp, and D-Adam at the representative setting $p = 0.3$, $q(t) = 1$, and $\eta_0 = 0.1$; these reference curves are not additional connectivity sweeps.

Synthetic online performance under varying connectivity. The ER curves sweep DECO-**ii** (KT) over $p \in \{0.1, 0.3, 1.0\}$; D-AdaGrad, D-RMSProp, and D-Adam are $p = 0.3$, $q(t) = 1$, $\eta_0 = 0.1$ references. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

The figure also includes D-AdaGrad, D-RMSProp, and D-Adam with $q(t) = 1$ and $\eta_0 = 0.1$ as online adaptive reference curves; only DECO is varied across gossip schedules.

Cumulative network loss on the synthetic online stream: DECO-**ii** (KT) is shown with constant, logarithmic, and linear gossip schedules; D-AdaGrad, D-RMSProp, and D-Adam use $q(t) = 1$ and $\eta_0 = 0.1$; increasing DECO's communication budget reduces loss, while the adaptive curves provide same-communication references.

Final cumulative network loss on four online LIBSVM streams. Each panel shows the complete 25-value η_0 sweep for DOGD, D-AdaGrad, D-RMSProp, and D-Adam on a log-log scale. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^3]$. Table II reports the best cumulative network loss in hindsight over the displayed grid, together with the selected η_0 and total communicated scalars.

Online LIBSVM experiments with $N = 20$ and $q(t) = 1$. For DOGD and the three adaptive baselines, "Selected η_0" is the best value in hindsight among the 25 values plotted in Fig. 6; it is not additional tuning hidden from the figure. Lower cumulative network loss is better.

Dataset	T	Method	Selected η_0	Cumulative network loss	Scalars
space_ga	155	DECO-i (KT)	—	92.51	43,400
space_ga	155	DECO-ii (KT)	—	101.76	37,200
space_ga	155	DOGD	1	88.81	37,200
space_ga	155	D-AdaGrad	0.316	89.21	37,200
space_ga	155	D-RMSProp	0.0316	89.28	37,200
space_ga	155	D-Adam	0.178	89.53	37,200
cpusmall	409	DECO-i (KT)	—	136.59	212,680
cpusmall	409	DECO-ii (KT)	—	147.13	196,320
cpusmall	409	DOGD	1.78	127.63	196,320
cpusmall	409	D-AdaGrad	0.562	121.73	196,320
cpusmall	409	D-RMSProp	0.0316	122.99	196,320
cpusmall	409	D-Adam	0.1	122.09	196,320
cadata	1,032	DECO-i (KT)	—	517.87	371,520
cadata	1,032	DECO-ii (KT)	—	583.69	330,240
cadata	1,032	DOGD	1.78	495.55	330,240
cadata	1,032	D-AdaGrad	1	493.68	330,240
cadata	1,032	D-RMSProp	0.0562	496.91	330,240
cadata	1,032	D-Adam	0.1	499.85	330,240
abalone	208	DECO-i (KT)	—	130.13	74,880
abalone	208	DECO-ii (KT)	—	137.66	66,560
abalone	208	DOGD	1.78	121.45	66,560
abalone	208	D-AdaGrad	0.562	121.30	66,560
abalone	208	D-RMSProp	0.0562	121.50	66,560
abalone	208	D-Adam	0.1	121.07	66,560

The best-in-grid adaptive baselines are competitive with or better than DECO on several streams, but the selected scale varies by method and dataset. The empirical conclusion is therefore not that DECO uniformly dominates tuned adaptive methods; rather, DECO avoids this learning-rate sweep while providing explicit network-regret guarantees in the convex online setting.

Finally, we evaluate DECO, DOGD, and the three online adaptive baselines on four real-world LIBSVM regression streams. The cycle graph, sample order, horizon, and one-gossip-round budget are identical for every method and every value of η_0. Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, as the best-in-grid values in Table II make explicit. The supported conclusion is that DECO removes learning-rate selection, not that it uniformly improves upon every tuned adaptive optimizer.

The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.

Comment 6

The discussion of prior works on parameter-free decentralized learning is insufficient.

Response: Thank you for the constructive comment.

We expanded the related-work positioning to include the recent parameter-free decentralized optimization papers named by the reviewer. These methods optimize fixed stochastic, strongly convex smooth, or composite objectives and are evaluated by convergence to a minimizer; they do not implement the same sequential predict-before-feedback protocol or optimize network regret. We therefore do not present them as directly interchangeable experimental baselines. Instead, the empirical comparison uses D-AdaGrad, D-RMSProp, and D-Adam under exactly the same causal decentralized online protocol as DOGD and DECO. The relevant added manuscript text (with its citations retained in the manuscript) is:

> Recent parameter-free decentralized optimization methods address nonconvex stochastic objectives [17], strongly convex smooth objectives [18], and composite convex objectives [19]. These works study convergence to a minimizer under their respective optimization models, whereas our setting considers sequential convex losses and network regret against a comparator. The problem models and performance criteria are therefore complementary rather than directly interchangeable.
> All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^{3}]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam.

Comment 7

The parameter-free benefit is restricted to convex online learning; many decentralized learning problems are nonconvex.

Response: Thank you for the constructive comment.

We added this as an explicit limitation. Our regret guarantees are for convex online learning (and the associated linearized OLO analysis). Extending the approach to nonconvex decentralized learning would require a different performance criterion and is outside the scope of the present paper. The manuscript now states:

Our guarantees are restricted to convex online learning and its linearized OLO reduction. They do not establish stationarity or convergence for nonconvex decentralized learning, which requires a different performance criterion. Extending parameter-free decentralized coin betting to that setting is outside the present scope.

Comment 8

The communication design of DECO-i should be justified, and DECO-i and DECO-ii should be compared under normalized communication cost.

Response: Thank you for the constructive comment.

We now justify DECO-i as the direct decentralized form of classical wealth-based coin betting and DECO-ii as the one-state betting-function variant that avoids communicating wealth. The implementation gossips only G for DECO-ii and (Wealth, G) for DECO-i. The rerun reports both variants under the same cycle graph, data order, horizon, and gossip schedule on all four LIBSVM streams. For example, on `space_ga`, DECO-i communicates 43,400 scalars and DECO-ii communicates 37,200; on `cadata`, the corresponding counts are 371,520 and 330,240. Table II reports the loss and scalar count for every method so the update-rule and communication-state differences are visible together. The exact added manuscript text and complete communication-normalized table are:

DECO-**i** is the direct decentralized analogue of classical wealth-based coin betting: because its decision depends on both $\mathrm{Wealth}_{n,t-1}$ and $G_{n,t-1}$, both states are gossiped. DECO-**ii** is the one-state betting-function variant $x_{n,t} = h_t(G_{n,t-1})$ and gossips only G. Thus, DECO-**i** is a reference construction that preserves the classical wealth recursion, whereas DECO-**ii** reduces the communicated state. We report total transmitted scalars so that their empirical comparison makes this difference explicit.

If d is the model dimension and $|E_{\mathrm{dir}}|$ is the number of directed non-self neighbor transmissions, one gossip round communicates $|E_{\mathrm{dir}}|d$ real scalars for DECO-**ii**, DOGD, and the adaptive decentralized baselines, and $|E_{\mathrm{dir}}|(d+1)$ real scalars for DECO-**i** because it additionally communicates the scalar wealth. Over a horizon T, a schedule $q(t)$ therefore communicates $|E_{\mathrm{dir}}|d\sum_{t=1}^{T}q(t)$ scalars for one-vector methods and $|E_{\mathrm{dir}}|(d+1)\sum_{t=1}^{T}q(t)$ scalars for DECO-**i**.

Online LIBSVM experiments with $N = 20$ and $q(t) = 1$. For DOGD and the three adaptive baselines, "Selected η_0" is the best value in hindsight among the 25 values plotted in Fig. 6; it is not additional tuning hidden from the figure. Lower cumulative network loss is better.

Dataset	T	Method	Selected η_0	Cumulative network loss	Scalars
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